

Real-time Toilet Paper Remaining Level Monitoring System

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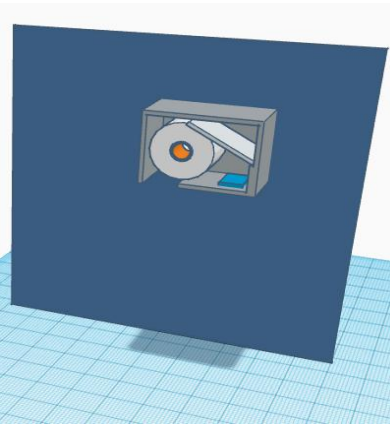
- **Problems**

- Toilet paper availability is critical in high-traffic public restrooms
- Manual inspection makes efficient restroom management difficult in multi-stall environments

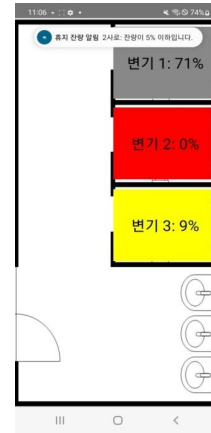
- **Solutions**



< Dispenser >



< Installation Concept >

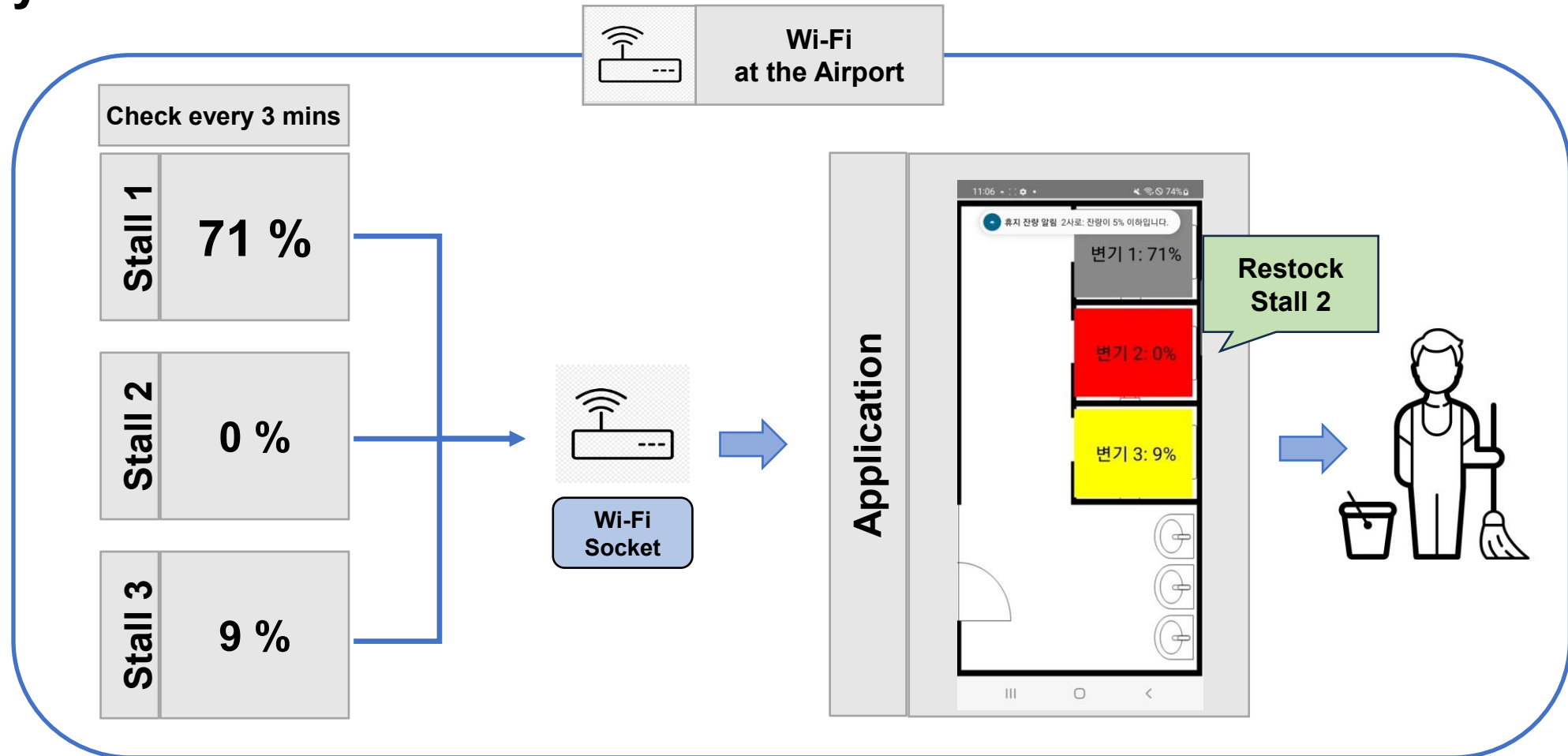


< Application >

- Arduino-based smart toilet paper dispenser for level detection
- Wi-Fi-based monitoring and management application

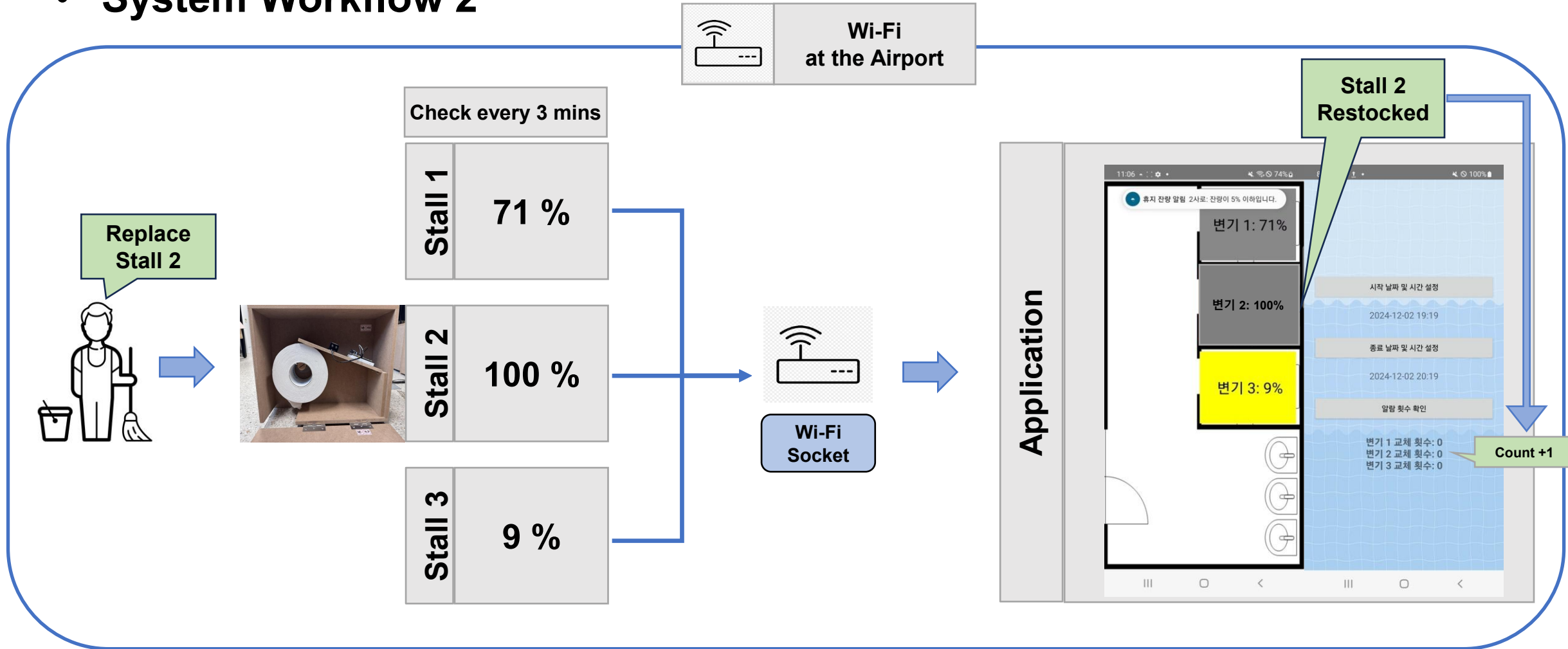
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- System Workflow 1



Real-time Toilet Paper Remaining Level Monitoring System

- System Workflow 2



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- **Results**

2024.09 ~ 2024.12

- IMU-based toilet paper level detection dispenser
- Real-time monitoring application
- Competition awards & Patent pending

- **My Contribution**

- Application UI development & sensor data visualization
- IMU sensor data processing logic
- Replacement event detection
- Team Leader
- Poster presentations at competitions



< Competition >